

Containerized Pipeline: Quick Start Guide

A beginner-friendly manual for accessing the research environment | Prepared by Devraj Singh

This document explains how to take a pre-packaged scientific environment (a Docker Image) and run it on any computer. You do not need to know how to code to follow these steps. By the end of this guide, you will have a fully functioning Linux environment loaded with bioinformatics tools (PLINK, BCFtools, LDetect) running safely inside your regular computer.

Phase 1: Installing Docker Desktop

Docker Desktop is the "player" that reads the environment file. You must install this first.

Windows Setup Steps

1. Go to **docker.com/products/docker-desktop** and download the Windows installer.
2. Run the installer. Ensure the option **"Use WSL 2 instead of Hyper-V"** is checked if prompted.
3. Once installed, your computer will likely restart.
4. Open the **Docker Desktop** app from your Start Menu. Accept the terms.
5. Wait until the indicator in the bottom-left corner turns green and says "Engine Running". Keep this app open in the background.

Mac Setup Steps

1. Go to **docker.com/products/docker-desktop** and download the Mac installer (choose Apple Silicon if you have an M1/M2/M3 chip, or Intel if you have an older Mac).
2. Open the downloaded .dmg file and drag the Docker icon into your Applications folder.
3. Open **Docker** from your Launchpad or Applications folder.
4. Grant it the required permissions (it will ask for your Mac password to install networking tools).
5. Wait for the Docker icon in your top menu bar to show a green dot indicating the engine is running.

Phase 2: Transferring the File

You will be provided with a compressed image file containing the pipeline. We will use the example filename: `my_research_pipeline.tar.gz`.

CRITICAL STEP: Never run the file directly from a USB drive or a cloud-synced folder (like OneDrive or iCloud Desktop). Large files get corrupted this way. Always copy the `.tar.gz` file directly into your local **Downloads** folder before proceeding.

Phase 3: Loading the Image into Docker

Now, we will "unpack" the file so Docker recognizes it.

Windows Loading the File

1. Open **Windows PowerShell** (Search for it in the Start Menu).
2. Run the following command to load the image. Replace Username with your actual Windows username:

```
docker load -i "C:\Users\Username\Downloads\my_research_pipeline.tar.gz"
```

Mac Loading the File

1. Open the **Terminal** app (Search for it using Spotlight/Cmd+Space).
2. Run the following command:

```
docker load -i ~/Downloads/my_research_pipeline.tar.gz
```

Wait for the process to finish. It will take a few minutes and will output "Loaded image: ref-panel-pipeline" when complete.

Phase 4: Running the Pipeline Environment

Your container is an isolated box. To allow it to read your data files (like `.vcf` files), you must create a bridge between a folder on your computer and the container.

1. Create a folder on your Desktop named **ProjectData**. Place any data files you want to analyze inside this folder.
2. Open your command line (PowerShell for Windows, Terminal for Mac) and run the startup command below.

Windows

Startup Command

```
docker run -it -v "C:\Users\Username\Desktop\ProjectData:/data" ref-panel-pipeline
```

Mac

Startup Command

```
docker run -it -v ~/Desktop/ProjectData:/data ref-panel-pipeline
```

What this command does:

- `run -it`: Starts the container and connects your keyboard to its terminal.
- `-v . . . : /data`: Creates a magic bridge. Anything you put in your "ProjectData" folder instantly appears in the "/data" folder inside Linux.
- `ref-panel-pipeline`: The name of the image you are starting.

Phase 5: Success & Verification

If successful, your terminal prompt will change to something like `root@a1b2c3d4: /app#`. You are now inside the Linux environment!

You can verify the tools are installed and ready by typing the following commands inside the container:

```
plink1.9 --version  
bcftools --version  
python3 -c "import ldetect; print('Success')"
```

To access the files you put in your Desktop folder, simply type `cd /data`.